Name: _____

TOPIC TEST 1

Chapter 8: Reducing balance loans and annuities

- Time allowed: 45 minutes
- CAS calculators may be used.
- Part A: 5 multiple-choice questions (5 marks)
- Part B: 5 short-answer questions (15 marks)
- Part C: 1 analysis question (18 marks)

Part A

Multiple-choice questions

5 multiple-choice questions
1 mark each: 5 marks
Circle the correct answer.

1 Elizabeth borrows $30 000 from the bank to buy a new car. Interest is charged on this loan at a rate of 8.75% p.a., calculated monthly. If Elizabeth pays $500 a month towards repaying the loan, how much does she owe the bank after 5 years?

A $569.32

B $0.00

C $83 856.84

D $2625

E $8925.58

[1 mark]

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2 Adrian takes out a loan for $200 000 at 4.65% p.a., calculated monthly, to be paid back over a period of 30 years. If, after 10 years, Adrian increases his monthly repayments to $1500, how many months earlier will he pay off the loan?

A 101

B 139

C 172

D 189

E 259

[1 mark]

3 Anushka receives an inheritance of $0.5 million. She invests the money at 5.75% p.a., compounding monthly. If Anushka chooses to receive monthly payments of $4000, how long will the money last (to the nearest month)?

A 10 years 5 months

B 15 years 11 months

C 10 years 8 months

D 11 years 0 months

E 7 years 2 months

[1 mark]

4 Alistair receives an inheritance of $250 000 and invests in a perpetuity that pays him 4.2% interest p.a., compounding monthly. How much can Alistair expect to receive each month from this perpetuity?

A $10 500

B $875

C $21 310.33

D $41 745.65

E $50 0875

[1 mark]

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5 Annabelle opens a savings account with an interest rate of 6.25%. She begins with a $15 000 investment and deposits $1000 per month. What is the amount of interest earned over 5 years?

A $20 447.10

B $90 706.09

C $5447.10

D $447.10

E $15 706.09

[1 mark]

Part B

Short-answer questions

5 short-answer questions
15 marks
Show your working where appropriate.

1 Jack took out a loan for $25 000. He makes monthly repayments of $600 and interest is calculated at a rate of 9.75% p.a., adjusted monthly.

a Write a difference equation that gives the amount still owing on the loan after n months.

b How much does Jack owe after 3 years?

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c How much interest has Jack paid in this time?

d At this rate, how long until Jack pays off the loan?

[4 marks]

2 Emily borrows $45 000 from a bank at 10.2% p.a. interest, adjusted monthly. If the loan is to be repaid after 12 years, find the:

a monthly repayment

b amount of interest paid.

[2 marks]

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3 Lachlan takes out a loan for $20 000 at 7.8% p.a. interest, adjusted quarterly.

a How many years until the loan is repaid if Lachlan repays $987.96 per quarter?

b If Lachlan needs to pay off the loan in 4 years, find the quarterly repayment required.

c How much interest would Lachlan save by repaying according to part **b** rather than part **a**?

[4 marks]

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4 Mustafa purchases a $150 000 annuity. Interest is paid at 6.75% p.a. compounded quarterly.

a If he wants the annuity to last 10 years, what would be the quarterly payments from the bank?

b If Mustafa decides to receive payments of $4000 per quarter, how long would the annuity last?

[2 marks]

5 Rebecca wants to set up a perpetuity. She decides to invest in stocks that return 7.4% p.a., compounded monthly.

a If she invests $100 000, how much can she expect to receive each month?

b If Rebecca wants to receive $1000 each month, how much more will she need to invest?

[3 marks]

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Part C

Analysis question

1 analysis question
18 marks
Show your working where appropriate.

1 Cameron and Sarah are to share equally in an inheritance of $450 000. Sarah decides to invest her money, whereas Cameron uses it as a deposit for a house loan.

Sarah invests her money in an account at 4.99% p.a., compounding monthly.

a If she leaves this untouched (no deposits or withdrawals) for 5 years, what would be her account balance?

b Sarah decides to deposit $500 a month into the account. How much would she have after 5 years?

After 3 years, the interest rate drops to 4.59% p.a.

c How much less money would Sarah have after a total of 5 years if she continued depositing $500 a month?

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d To what amount should Sarah raise her monthly deposits in order to ensure her bank balance reaches the 5-year total calculated in part **b**?

Once the interest rate dropped, Sarah decided that rather than increasing her monthly deposit, she would change her investment to shares earning 5.25% p.a. compounding quarterly and set up a perpetuity.

e How much should Sarah expect to receive quarterly from her perpetuity?

f If Sarah decided that she would instead prefer to receive quarterly payments of $6000, how long would this annuity now last?

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With his share of the money, Cameron places a deposit on a $550 000 house and takes out a loan for the remainder of the house value.

g If he takes out a 25-year loan at 5.15% p.a. compounding monthly, what monthly repayments would he need to make?

After 10 years, the interest rate dropped to 4.25% p.a.

h If Cameron continued with his monthly repayments, how much:

i time would he save on his loan and

ii how much interest would he save on his loan?

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i If Cameron instead chose to keep the 25-year term, to what amount would his monthly repayments drop?

j Find the total amount of interest Cameron would pay if he chose to keep the 25-year term.

k Would this amount differ from the amount of interest he would have paid if the interest rate had remained the same? If so, by how much?

l What would be your recommendation to Cameron?

[18 marks]

[Total: 38 marks]

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Answers

Part A – Multiple-choice questions **[1 mark each]**

1 Using:

$N = 5 \times 12 = 60$

$I\% = 8.75$

$PV = 30\,000$

$PMT = -500$

$P/Y = C/Y = 12$

gives an FV of $-\$8925.58$

$\therefore$ **E**

2 Using:

$N = 30 \times 12 = 360$

$I\% = 4.65$

$PV = 200\,000$

$P/Y = C/Y = 12$

$FV = 0$

gives a PMT of $-\$1031.27$

Next, using:

$N = 10 \times 12 = 120$

$I\% = 4.65$

$PV = 200\,000$

$P/Y = C/Y = 12$

PMT of $-\$1031.27$

gives an FV of \$160 941.64, which is the amount he still owes after 10 years.

Next, using:

$I\% = 4.65$

$PV = 160\,941.64$

$P/Y = C/Y = 12$

PMT of $-\$1500$

$FV = 0$

gives $N = 138.97$, which is the number of months until he pays off the loan.

139 months (rounded) + 120 months = 259 months in total to pay off the loan.

360 months − 259 months = 101 months saved by increasing payments.

$\therefore$ **A**

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3 Using:

$I\% = 5.75$

$PV = -500\,000$

$PMT = \$4000$

$P/Y = C/Y = 12$

$FV = 0$

gives $N = 191.14$

$\frac{191}{12} = 15.9275$ years, which is 15 years 11 months.

$\therefore$ **B**

4 Using the formula $Q = \frac{Pr}{100}$, where $P = 250\,000$ and $r = \frac{4.2}{12}$, we find $Q = \$875$.

$\therefore$ **B**

5 Using:

$N = 5 \times 12 = 60$

$I\% = 6.25$

$PV = -15\,000$

$PMT = -\$1000$

$P/Y = C/Y = 12$

gives $FV = \$90\,706.09$

$\$90\,706.09 - (15\,000 + 1000 \times 60) = \$15\,706.09$

$\therefore$ **E**

Part B – Short-answer questions

1 a $A_n = \left(1 + \frac{r}{100}\right)A_{n-1} - \text{monthly repayment}$

$r = \frac{9.75}{12}$ (interest rate per month)

Monthly repayment $= 600$

Initial balance $= A_0 = 25\,000$

$A_n = \left(1 + \frac{9.75}{1200}\right)A_{n-1} - 600$, where $A_0 = 25\,000$ **[1 mark]**

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b Using:

$N = 3 \times 12 = 36$

$I\% = 9.75$

$PV = 25\,000$

$PMT = -600$

$P/Y = C/Y = 12$

gives $FV = \$8480.71$

After 3 years, Jack owes \$8480.71. **[1 mark]**

c Jack has paid $\$600 \times 36 = \$21\,600$.

Jack has paid off $\$25\,000 - \$8480.71 = \$16\,519.29$.

Amount of interest $= \$21\,600 - \$16\,519.29 = \$5080.71$. **[1 mark]**

d Using:

$I\% = 9.75$

$PV = 25\,000$

$PMT = -600$

$FV = 0$

$P/Y = C/Y = 12$

gives $N = 51.08$ months

At this rate, Jack pays off the loan in 4 years, 3 months. **[1 mark]**

2 a Using:

$N = 12 \times 12 = 144$

$I\% = 10.2$

$PV = 45\,000$

$FV = 0$

$P/Y = C/Y = 12$

gives $PMT = -\$543.00$

The monthly repayment is \$543. **[1 mark]**

b Amount of interest paid = total amount repaid − value of loan

$= 543 \times 144 - 45\,000 = \$33\,192$.

Emily paid \$33 192 in interest. **[1 mark]**

3 a Using:

$I\% = 7.8$

$PV = 20\,000$

$PMT = -\$987.96$

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$FV = 0$

$P/Y = C/Y = 4$

gives $N = 26$ quarters

It therefore takes 26 quarters to fully repay the loan, which is $\frac{26}{4} = 6.5$ years. **[1 mark]**

b Using:

$N = 4 \times 4 = 16$

$I\% = 7.8$

$PV = 20\,000$

$FV = 0$

P/Y = C/Y = 4

gives $PMT = -\$1467.17$

Lachlan would need to pay \$1467.17 per quarter to pay off the loan in 4 years. **[1 mark]**

c Interest paid in part **a** = total amount repaid − value of loan

$= 987.96 \times 26 - 20\,000 = \$5686.96.$

Interest paid in part **b** = total amount repaid − value of loan

$= 1467.17 \times 16 - 20\,000 = \$3474.72.$

Interest saved = \$5686.96 − \$3474.72 **[1 mark]**

= \$2212.24.

Lachlan would save \$2212.24 in interest if he chose to repay according to part **b** rather than part **a**. **[1 mark]**

4 a Using:

$N = 10 \times 4 = 40$

$I\% = 6.75$

$PV = -150\,000$

$FV = 0$

$P/Y = C/Y = 4$

gives $PMT = -\$5187.33$

Mustafa would receive quarterly payments of \$5187.33. **[1 mark]**

b $I\% = 6.75$

$PV = -150\,000$

gives $PMT = -\$4000$

$FV = 0$

$P/Y = C/Y = 4$

gives $N = 59.87$ quarters

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This is $\frac{60}{4} = 15$ years.

Mustafa's annuity would last for 15 years. **[1 mark]**

5 a Using the formula $Q = \frac{Pr}{100}$, where $P = 100\,000$ and $r = \frac{7.4}{12}$ gives $Q = \$616.67$.

Rebecca would receive \$616.67 per month. **[1 mark]**

b Using the formula $Q = \frac{Pr}{100}$, where $Q = 1000$ and $r = \frac{7.4}{12}$ gives $P = \$162\,162.16$. **[1 mark]**

Rebecca will need to invest \$162 162.16 − \$100 000 = \$62 162.16 more in order to receive monthly payments of \$1000. **[1 mark]**

Part C – Analysis question

1 a Compound interest formula: $A = P(1 + \frac{r}{100})^n$

$P = 225\,000$, $r = \frac{4.99}{12}$, $n = 12 \times 5 = 60$

$A = 225\,000(1 + \frac{4.99}{1200})^{60} = \$288\,611.96$

Sarah's account balance would be \$288 611.96. **[1 mark]**

b $N = 5 \times 12 = 60$

$I\% = 4.99$

$PV = -225\,000$

$PMT = -\$500$

$P/Y = C/Y = 12$

gives $FV = -\$322\,606.33$

If she deposited \$500 per month, Sarah would have \$322 606.33 after 5 years. **[1 mark]**

c $N = 3 \times 12 = 36$

$I\% = 4.99$

$PV = -225\,000$

$PMT = -\$500$

$P/Y = C/Y = 12$

gives $FV = -\$280\,626.97$

After 3 years, Sarah has \$280 626.97. **[1 mark]**

Next, using:

$N = 2 \times 12 = 24$

$I\% = 4.59$

$PV = -280\,626.97$

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$PMT = -\$500$

$P/Y = C/Y = 12$

gives $FV = -\$320\ 097.10$

After 5 years, Sarah has \$320 097.10. **[1 mark]**

As a result of the interest rate drop, Sarah would have:

\$322 606.33 − \$320 097.10 = \$2509.23 less money after 10 years. **[1 mark]**

d After 3 years, Sarah has \$280 626.97.

Using:

$N = 2 \times 12 = 24$

$I\% = 4.59$

$PV = -280\ 626.97$

$FV = -322\ 606.33$

$P/Y = C/Y = 12$

gives $PMT = -\$600.03$

Sarah should lift her monthly deposits to \$600.03 in order to ensure her bank balance reaches the previous 5-year total (\$322 606.33). **[1 mark]**

e Using the formula $Q = \frac{Pr}{100}$, where $P = 280\ 626.97$ and $r = \frac{5.25}{4}$
gives $Q = \$3683.23$. Therefore, Sarah should expect to receive \$3683.23 quarterly from her perpetuity.
[1 mark]

f Using:

$I\% = 5.25$

$PV = -280\ 626.97$

$PMT = -\$6000$

$FV = 0$

$P/Y = C/Y = 4$

gives $N = 72.97$

Therefore, the annuity would last 72.97 quarters, which is roughly 18.25 years.

[1 mark]

g Loan is for \$550 000 − \$225 000 = \$325 000

$N = 25 \times 12 = 300$

$I\% = 5.15$

$PV = 325\ 000$

$FV = 0$

$P/Y = C/Y = 12$

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gives $PMT = -1928.43$

Cameron will need to make monthly repayments of \$1928.43. **[1 mark]**

h i $N = 10 \times 12 = 120$

$I\% = 5.15$

$PV = 325\,000$

$PMT = -1928.43$

$P/Y = C/Y = 12$

gives $FV = -241\,467.25$

Cameron will still owe \$241 467.25 after 10 years. **[1 mark]**

Next, using:

$I\% = 4.25$

$PV = 241\,467.25$

$PMT = -1928.43$

$FV = 0$

$P/Y = C/Y = 12$

gives $N = 165.76$ months **[1 mark]**

Time saved = original time − actual time

= 300 months − (120 + 166) months = 14 months.

This is ≈ 1 year and 2 months.

Therefore, Cameron will save approximately 1 year and 2 months on his loan. **[1 mark]**

ii Interest saved = original interest to be paid − new interest paid

$= 1928.43 \times 300 - 1928.43 \times (120 + 166)$

$= 578\,529 - 551\,530.98$

= \$26 998.02 **[1 mark]**

i Cameron still owes \$241 467.25 after 10 years.

Using:

$N = 300 - 120 = 180$ **[1 mark]**

$I\% = 4.25$

$PV = 241\,467.25$

$FV = 0$

$P/Y = C/Y = 12$

gives $PMT = -1816.51$

Cameron's monthly repayments would drop to \$1816.51. **[1 mark]**

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j Interest paid = total amount repaid − value of loan

$= 1928.43 \times 120 + 1816.51 \times 180 - 325\,000 = \$233\,383.40.$ **[1 mark]**

k Original interest paid $= 1928.43 \times 300 - 325\,000 = \$253\,529.$

So, yes, this amount differs from the amount of interest Cameron would have paid if the interest rate had remained the same.

Therefore, Cameron saves $\$253\,529 - \$233\,383.40 = \$20\,145.60$ as a result of the interest rate cut. **[1 mark]**

l Cameron is best advised to maintain his monthly payments because he saves 1 year and 2 months in time, but, importantly, he saves:

$\$26\,998.02 - \$20\,145.60 = \$6852.42$ in interest. **[1 mark]**

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